

MA641 Research Project(Seasonal)

Yash Tanaji Bhosale

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```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.2  
  
##  
## Attaching package: 'lubridate'  
  
## The following objects are masked from 'package:base':  
##  
##   date, intersect, setdiff, union
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
# Load the CSV (replace with your correct filename/path)  
cpi_raw <- read.csv("CPIAUCSL.csv", stringsAsFactors = FALSE)  
  
# Convert date column (MM/DD/YYYY)  
cpi <- cpi_raw %>%  
  mutate(  
    Date = mdy(observation_date), # FIX: use mdy() not ymd()  
    CPI = as.numeric(CPIAUCSL) # ensure numeric  
  ) %>%  
  arrange(Date) %>%
```

```
select(Date, CPI)
```

```
# Inspect  
str(cpi)
```

```
## 'data.frame': 945 obs. of 2 variables:  
## $ Date: Date, format: "1947-01-01" "1947-02-01" ...  
## $ CPI : num 21.5 21.6 22 22 21.9 ...
```

```
head(cpi)
```

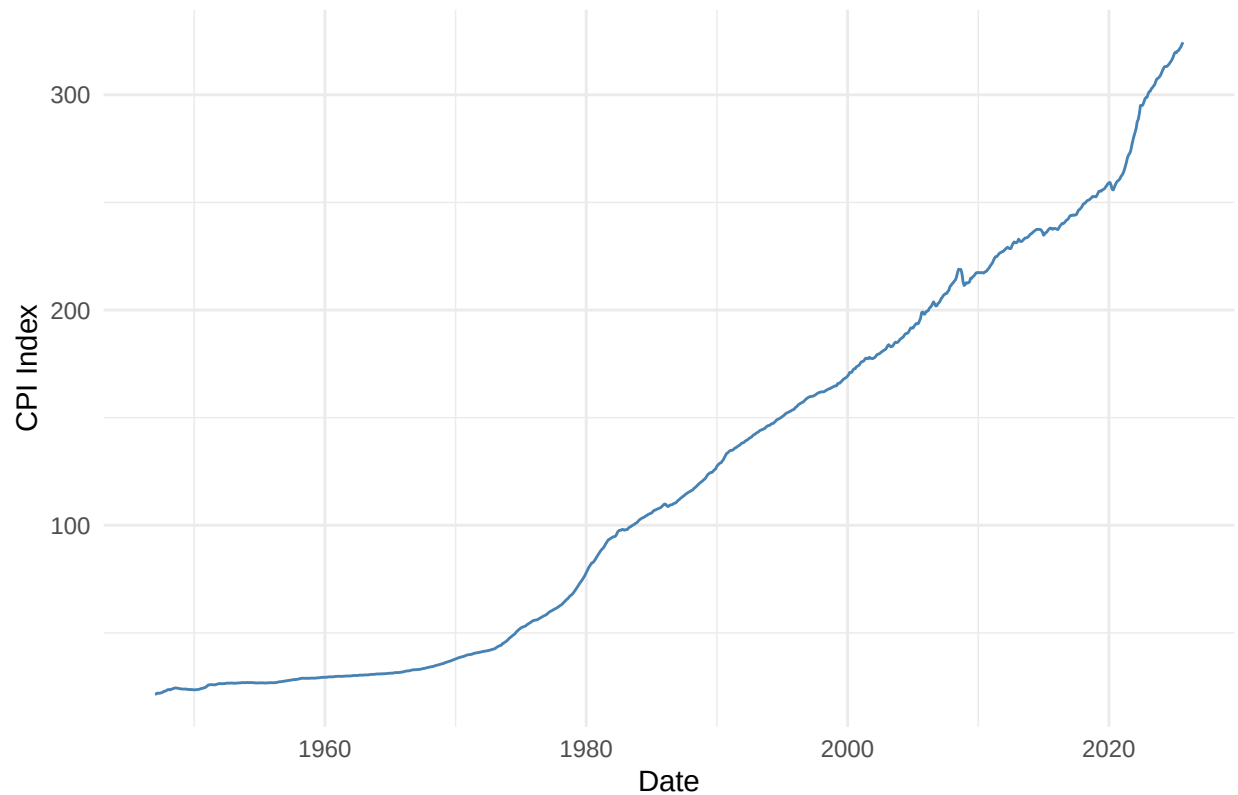
```
##      Date    CPI  
## 1 1947-01-01 21.48  
## 2 1947-02-01 21.62  
## 3 1947-03-01 22.00  
## 4 1947-04-01 22.00  
## 5 1947-05-01 21.95  
## 6 1947-06-01 22.08
```

```
tail(cpi)
```

```
##      Date    CPI  
## 940 2025-04-01 320.321  
## 941 2025-05-01 320.580  
## 942 2025-06-01 321.500  
## 943 2025-07-01 322.132  
## 944 2025-08-01 323.364  
## 945 2025-09-01 324.368
```

```
# Plot CPI  
ggplot(cpi, aes(x = Date, y = CPI)) +  
  geom_line(color = "steelblue") +  
  labs(  
    title = "U.S. CPI - Monthly",  
    x = "Date",  
    y = "CPI Index"  
  ) +  
  theme_minimal()
```

U.S. CPI — Monthly



```
## STEP 2A: Descriptive Statistics -----
```

```
library(moments)
```

```
## Warning: package 'moments' was built under R version 4.5.2
```

```
mean_cpi <- mean(cpi$CPI, na.rm = TRUE)
var_cpi  <- var(cpi$CPI, na.rm = TRUE)
sd_cpi   <- sd(cpi$CPI, na.rm = TRUE)
skew_cpi <- skewness(cpi$CPI, na.rm = TRUE)
kurt_cpi <- kurtosis(cpi$CPI, na.rm = TRUE)
```

```
cat("Mean:", mean_cpi, "\n")
```

```
## Mean: 123.4215
```

```
cat("Variance:", var_cpi, "\n")
```

```
## Variance: 7889.687
```

```
cat("SD:", sd_cpi, "\n")
```

```
## SD: 88.82391
```

```
cat("Skewness:", skew_cpi, "\n")
```

```
## Skewness: 0.4530693
```

```
cat("Kurtosis:", kurt_cpi, "\n")
```

```
## Kurtosis: 1.927122
```

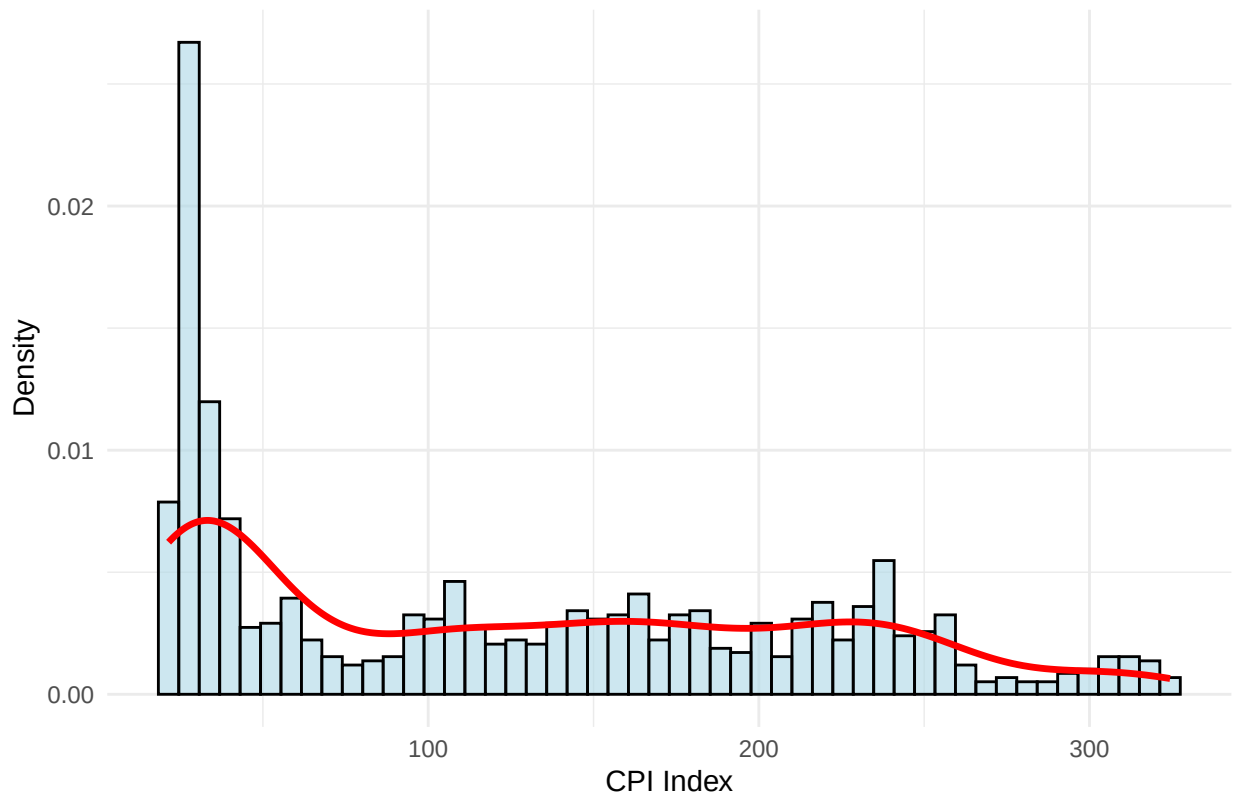
```
## STEP 2B: Histogram + Density -----
```

```
ggplot(cpi, aes(x = CPI)) +  
  geom_histogram(aes(y = ..density..),  
                 bins = 50,  
                 fill = "lightblue",  
                 color = "black",  
                 alpha = 0.6) +  
  geom_density(color = "red", size = 1.2) +  
  labs(title = "Histogram & Density of CPI",  
       x = "CPI Index",  
       y = "Density") +  
  theme_minimal()
```

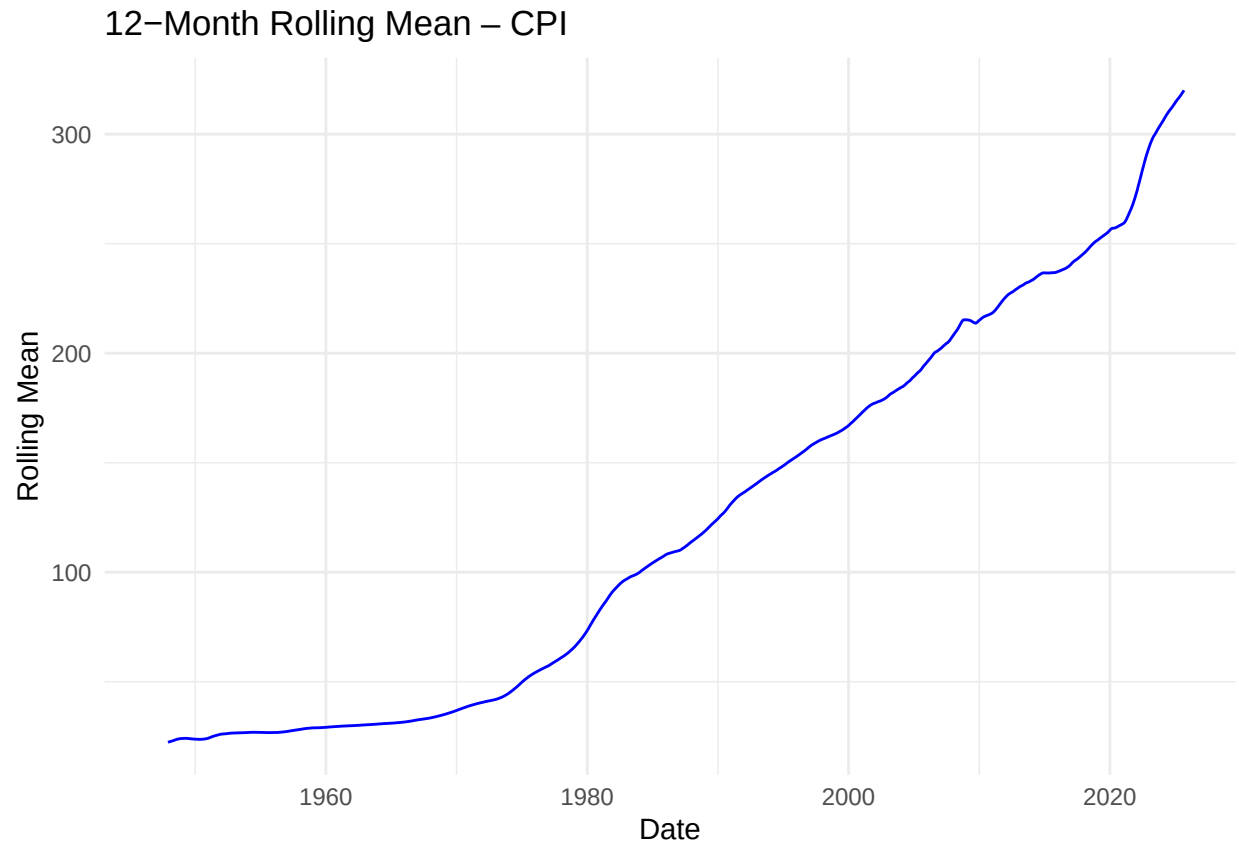
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use 'linewidth' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

```
## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.  
## i Please use 'after_stat(density)' instead.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

Histogram & Density of CPI



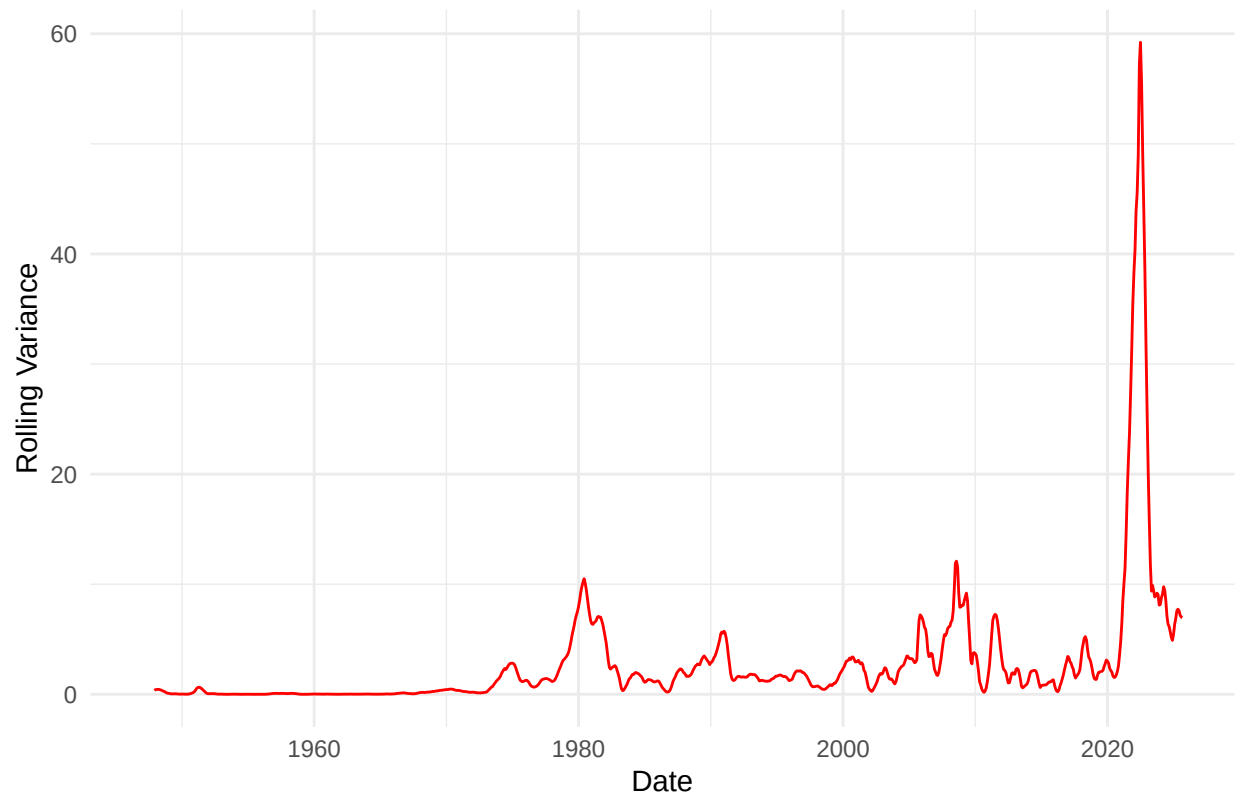
```
## STEP 2C: Rolling Mean & Variance -----  
  
library(zoo)  
  
##  
## Attaching package: 'zoo'  
  
## The following objects are masked from 'package:base':  
##  
##   as.Date, as.Date.numeric  
  
cpi$roll_mean <- rollmean(cpi$CPI, k = 12, fill = NA, align = "right")  
cpi$roll_var  <- rollapply(cpi$CPI, width = 12, FUN = var, fill = NA, align = "right")  
  
# Plot rolling mean  
ggplot(cpi, aes(x = Date, y = roll_mean)) +  
  geom_line(color = "blue") +  
  labs(title = "12-Month Rolling Mean - CPI",  
        y = "Rolling Mean") +  
  theme_minimal()  
  
## Warning: Removed 11 rows containing missing values or values outside the scale range  
## ('geom_line()').
```



```
# Plot rolling variance
ggplot(cpi, aes(x = Date, y = roll_var)) +
  geom_line(color = "red") +
  labs(title = "12-Month Rolling Variance - CPI",
        y = "Rolling Variance") +
  theme_minimal()
```

```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_line()').
```

12-Month Rolling Variance – CPI



```
## STEP 2D: ADF & KPSS tests -----
```

```
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.5.2
```

```
## Registered S3 method overwritten by 'quantmod':
```

```
##   method      from
```

```
## as.zoo.data.frame zoo
```

```
library(urca)
```

```
# ADF test
```

```
adf_cpi <- adf.test(cpi$CPI, alternative = "stationary")
```

```
adf_cpi
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: cpi$CPI
```

```
## Dickey-Fuller = -0.72277, Lag order = 9, p-value = 0.9685
```

```
## alternative hypothesis: stationary
```

```

# KPSS test (Level stationarity)
kpss_cpi <- ur.kpss(cpi$CPI, type = "mu")
summary(kpss_cpi)

##
## #####
## # KPSS Unit Root Test #
## #####
##
## Test is of type: mu with 7 lags.
##
## Value of test-statistic is: 11.5901
##
## Critical value for a significance level of:
##          10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463 0.574 0.739

## STEP 3: Make CPI stationary (Log + differencing + seasonal differencing) ----

library(dplyr)
library(ggplot2)
library(tseries)

# 3.1 Log transform
cpi$log_CPI <- log(cpi$CPI)

# 3.2 First difference (remove long-term trend)
cpi$diff_log_CPI <- c(NA, diff(cpi$log_CPI, lag = 1))

# 3.3 Seasonal difference (lag = 12)
cpi$season_diff <- c(rep(NA, 12), diff(cpi$log_CPI, lag = 12))

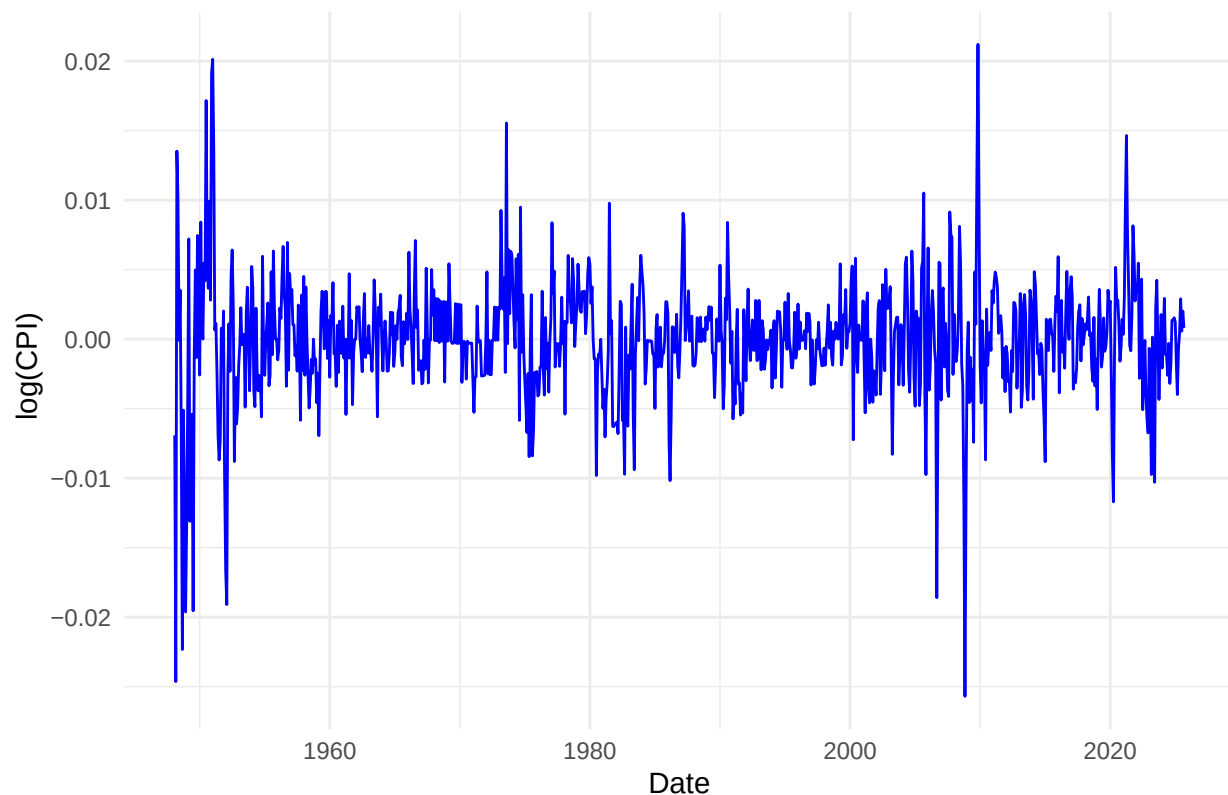
# 3.4 Combined difference: (d=1 AND D=1)
# log(CPI)
cpi$diff_season_diff <- c(rep(NA, 12), diff(cpi$diff_log_CPI, lag = 12))

# Clean dataset: remove NA rows
cpi_diff <- cpi %>%
  filter(!is.na(diff_season_diff))

# 3.5 Plot the twice-differenced series
ggplot(cpi_diff, aes(x = Date, y = diff_season_diff)) +
  geom_line(color = "blue") +
  labs(
    title = "Differenced & Seasonally Differenced Log CPI",
    y = " log(CPI)"
  ) +
  theme_minimal()

```


Differenced & Seasonally Differenced Log CPI



```
# 3.6 ADF test on transformed CPI
```

```
adf_cpi_transformed <- adf.test(cpi_diff$diff_season_diff, alternative = "stationary")
```

```
## Warning in adf.test(cpi_diff$diff_season_diff, alternative = "stationary"):
```

```
## p-value smaller than printed p-value
```

```
adf_cpi_transformed
```

```
##
```

```
## Augmented Dickey-Fuller Test
```

```
##
```

```
## data: cpi_diff$diff_season_diff
```

```
## Dickey-Fuller = -7.1008, Lag order = 9, p-value = 0.01
```

```
## alternative hypothesis: stationary
```

```
## STEP 4: Seasonal ACF & PACF for CPI -----
```

```
library(forecast)
```

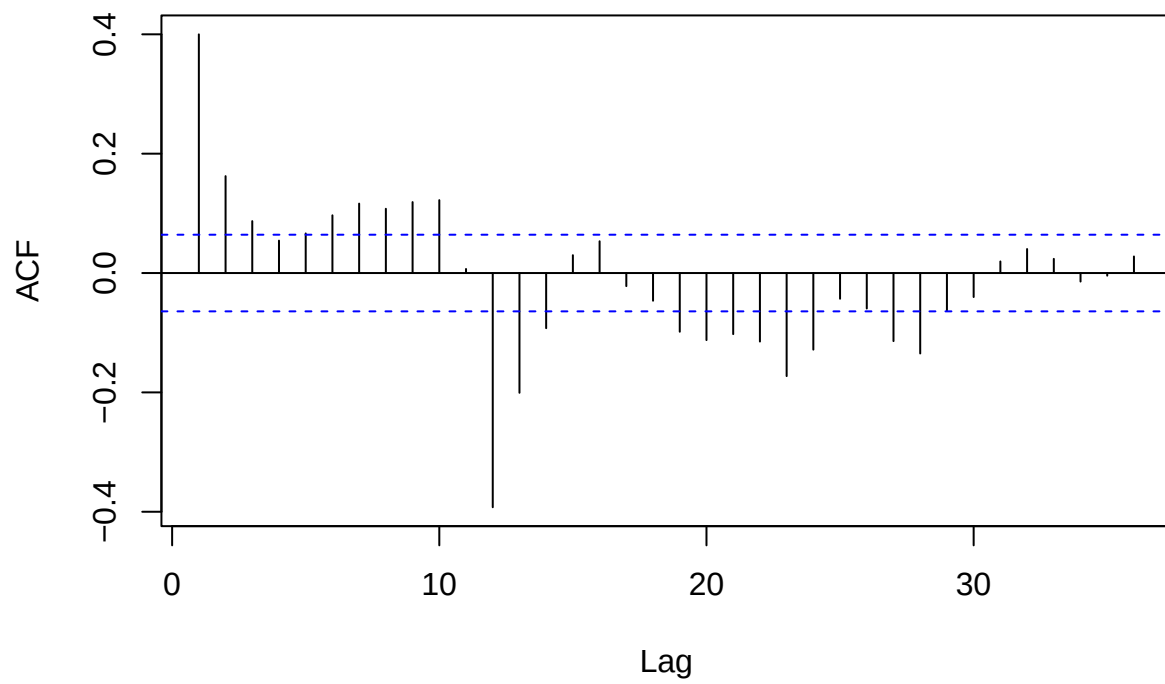
```
# Use the fully differenced series
```

```
y_cpi <- cpi_diff$diff_season_diff
```

```
# 4.1 ACF (show up to lag 36 to see seasonal spikes)
```

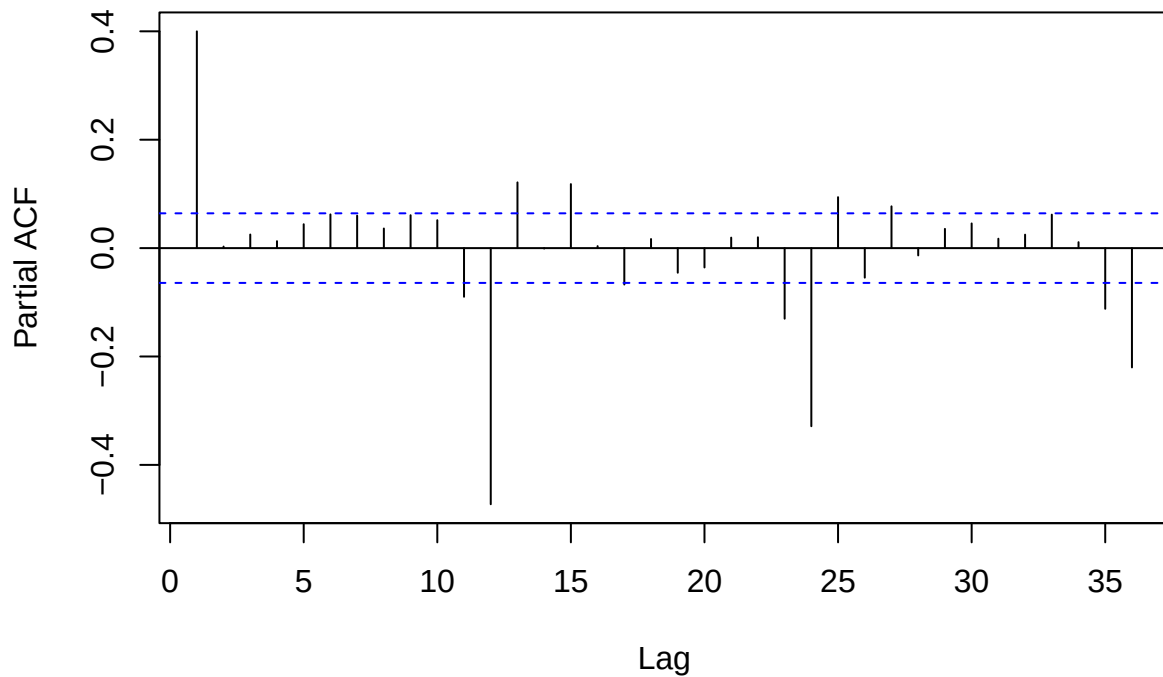
```
Acf(y_cpi, lag.max = 36, main = "ACF + Seasonal CPI")
```

ACF + Seasonal CPI



```
# 4.2 PACF  
Pacf(y_cpi, lag.max = 36, main = "PACF + Seasonal CPI")
```

PACF + Seasonal CPI



```
library(forecast)
library(dplyr)

y <- cpi$CPI

# Fit all candidate models -----

fit_112_011 <- Arima(y, order=c(1,1,2), seasonal=list(order=c(0,1,1), period=12))
fit_212_011 <- Arima(y, order=c(2,1,2), seasonal=list(order=c(0,1,1), period=12))
fit_111_111 <- Arima(y, order=c(1,1,1), seasonal=list(order=c(1,1,1), period=12))
fit_211_011 <- Arima(y, order=c(2,1,1), seasonal=list(order=c(0,1,1), period=12))
fit_011_111 <- Arima(y, order=c(0,1,1), seasonal=list(order=c(1,1,1), period=12))
fit_111_110 <- Arima(y, order=c(1,1,1), seasonal=list(order=c(1,1,0), period=12))
fit_110_110 <- Arima(y, order=c(1,1,0), seasonal=list(order=c(1,1,0), period=12))

# Build comparison table -----

model_comp <- tibble(
  model = c(
    "SARIMA(1,1,2)(0,1,1)[12]",
    "SARIMA(2,1,2)(0,1,1)[12]",
    "SARIMA(1,1,1)(1,1,1)[12]",
    "SARIMA(2,1,1)(0,1,1)[12]",
    "SARIMA(0,1,1)(1,1,1)[12]",
    "SARIMA(1,1,1)(1,1,0)[12]",
    "SARIMA(1,1,0)(1,1,0)[12]"
  )
)
```

```

),
AIC = c(
  AIC(fit_112_011),
  AIC(fit_212_011),
  AIC(fit_111_111),
  AIC(fit_211_011),
  AIC(fit_011_111),
  AIC(fit_111_110),
  AIC(fit_110_110)
),
BIC = c(
  BIC(fit_112_011),
  BIC(fit_212_011),
  BIC(fit_111_111),
  BIC(fit_211_011),
  BIC(fit_011_111),
  BIC(fit_111_110),
  BIC(fit_110_110)
)
)

```

```

# Print table
model_comp

```

```

## # A tibble: 7 x 3
##   model                AIC   BIC
##   <chr>              <dbl> <dbl>
## 1 SARIMA(1,1,2)(0,1,1)[12] 842. 866.
## 2 SARIMA(2,1,2)(0,1,1)[12] 842. 871.
## 3 SARIMA(1,1,1)(1,1,1)[12] 886. 910.
## 4 SARIMA(2,1,1)(0,1,1)[12] 889. 913.
## 5 SARIMA(0,1,1)(1,1,1)[12] 920. 940.
## 6 SARIMA(1,1,1)(1,1,0)[12] 1228. 1247.
## 7 SARIMA(1,1,0)(1,1,0)[12] 1259. 1273.

```

```

final_sarima <- Arima(y,
  order = c(1,1,2),
  seasonal = list(order=c(0,1,1), period = 12),
  include.constant = FALSE)

```

```

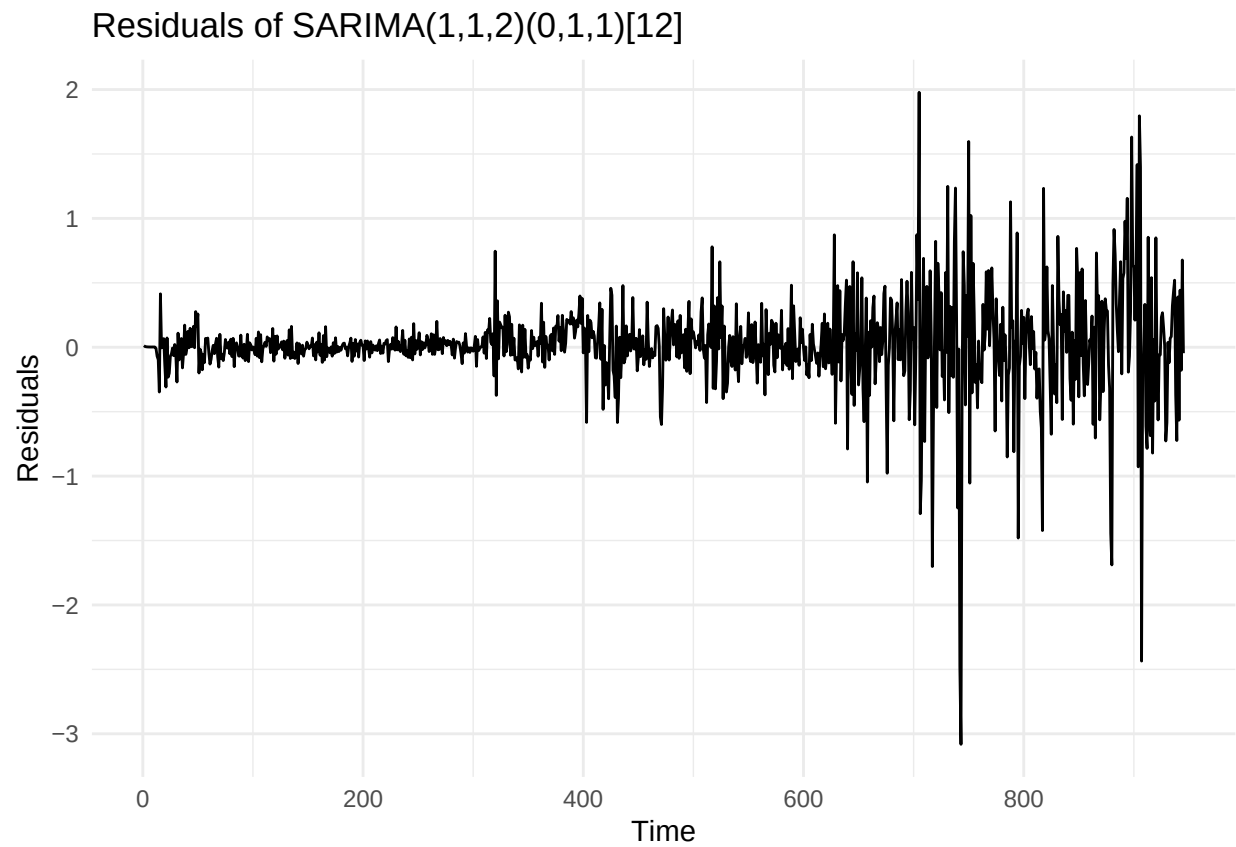
# STEP 6: Diagnostic Checking -----

```

```

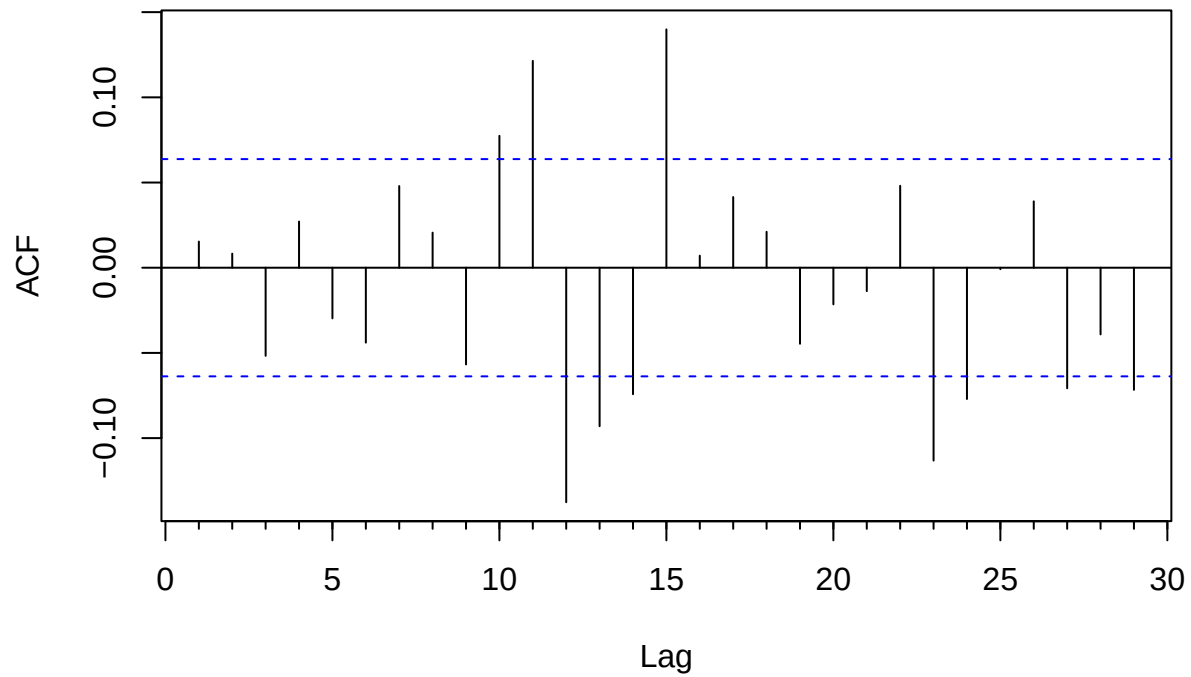
# 1. Residual time series plot
autoplot(residuals(final_sarima)) +
  ggtitle("Residuals of SARIMA(1,1,2)(0,1,1)[12]") +
  ylab("Residuals") +
  theme_minimal()

```



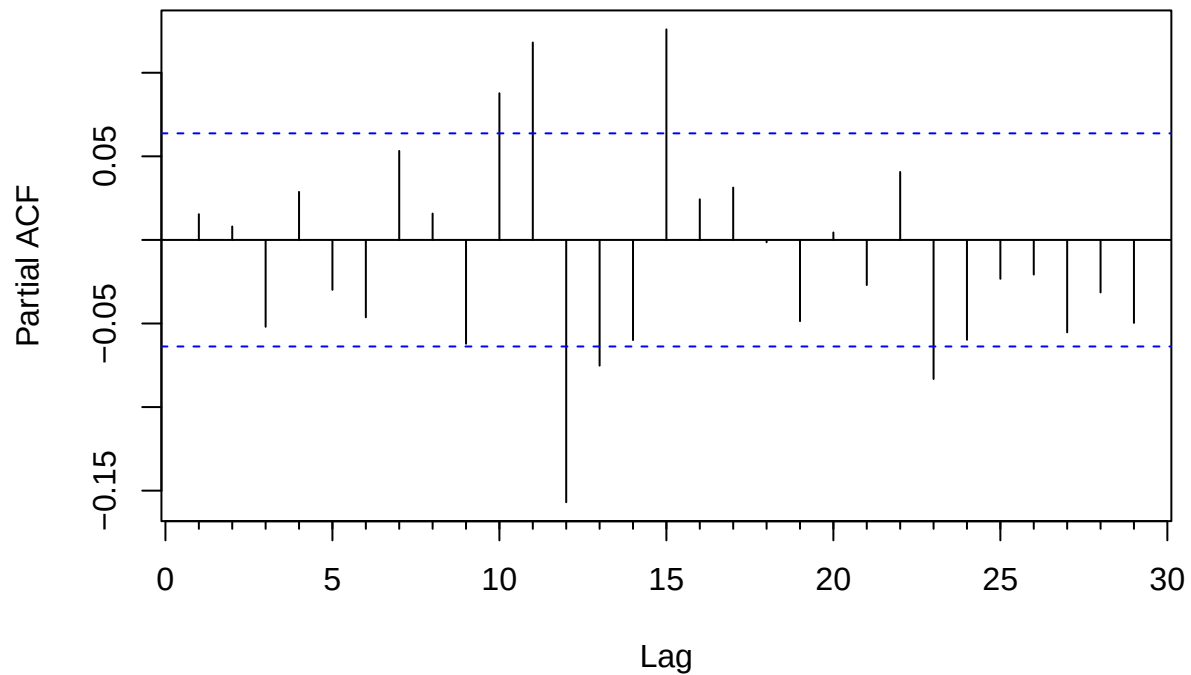
```
# 2. ACF of residuals
Acf(residuals(final_sarima),
    main = "ACF of Residuals - SARIMA(1,1,2)(0,1,1)[12]")
```

ACF of Residuals – SARIMA(1,1,2)(0,1,1)[12]



```
# 3. PACF of residuals (optional but useful)
Pacf(residuals(final_sarima),
      main = "PACF of Residuals - SARIMA(1,1,2)(0,1,1)[12]")
```

PACF of Residuals – SARIMA(1,1,2)(0,1,1)[12]



```
# 4. Ljung-Box test (lag = 24 recommended for monthly data)
```

```
Box.test(residuals(final_sarima),  
         lag = 24,  
         type = "Ljung-Box")
```

```
##
```

```
## Box-Ljung test
```

```
##
```

```
## data: residuals(final_sarima)
```

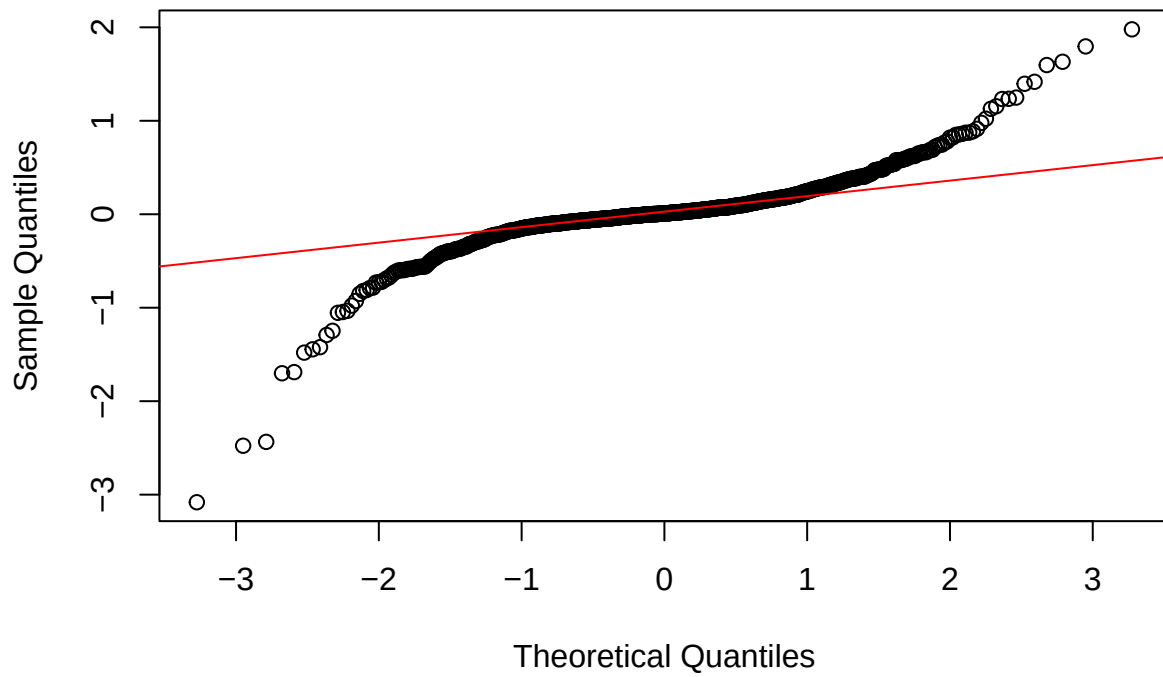
```
## X-squared = 107.49, df = 24, p-value = 1.541e-12
```

```
# 5. Normality check: QQ plot
```

```
qqnorm(residuals(final_sarima))
```

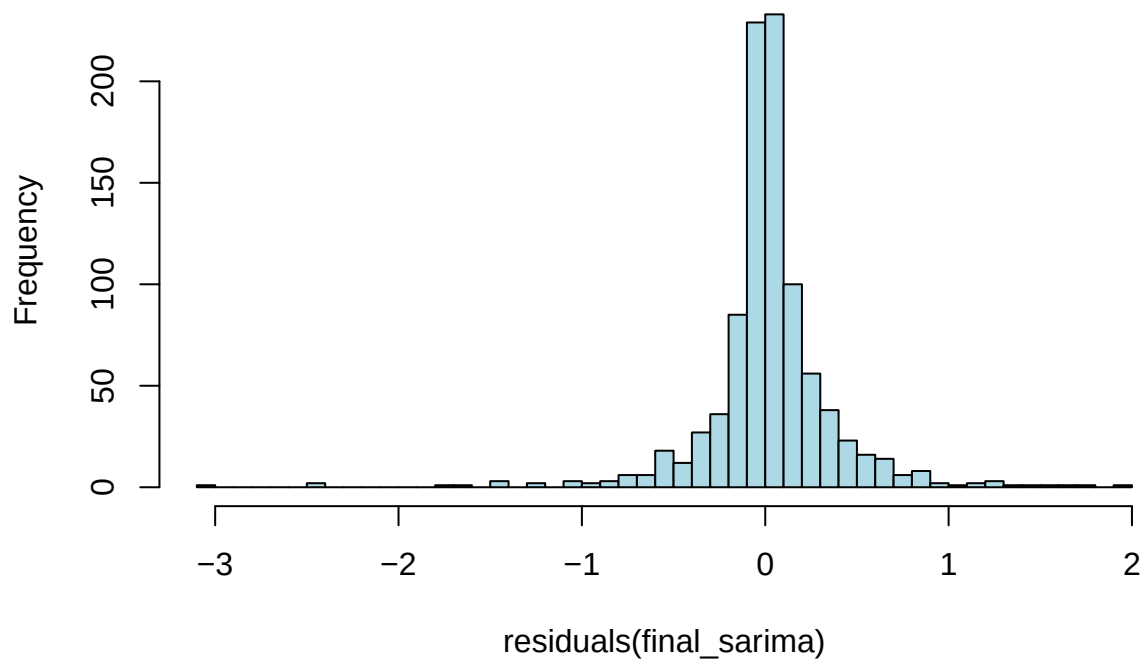
```
qqline(residuals(final_sarima), col = "red")
```

Normal Q-Q Plot



```
# 6. Histogram of residuals
hist(residuals(final_sarima),
     breaks = 40,
     main = "Histogram of Residuals",
     col = "lightblue",
     border = "black")
```


Histogram of Residuals



STEP 8: Forecast the next 12 months -----

```
library(forecast)
library(ggplot2)

# 12-month ahead forecast
cpi_forecast <- forecast(final_sarima, h = 12)

# Print forecast values
cpi_forecast
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 946	325.0029	324.5263	325.4794	324.2740	325.7317
## 947	325.5787	324.7008	326.4565	324.2361	326.9212
## 948	326.1808	324.9944	327.3673	324.3663	327.9954
## 949	326.9237	325.4596	328.3878	324.6846	329.1628
## 950	327.6377	325.9107	329.3647	324.9965	330.2789
## 951	328.2614	326.2793	330.2435	325.2300	331.2927
## 952	328.9015	326.6685	331.1344	325.4865	332.3165
## 953	329.5257	327.0442	332.0072	325.7305	333.3208
## 954	330.2868	327.5578	333.0157	326.1132	334.4604
## 955	330.8994	327.9233	333.8754	326.3479	335.4508
## 956	331.5518	328.3286	334.7750	326.6224	336.4812
## 957	332.2483	328.7776	335.7189	326.9404	337.5562

```
# Plot forecast
autoplot(cpi_forecast) +
  ggtitle("12-Month Forecast - CPI using SARIMA(1,1,2)(0,1,1)[12]") +
  xlab("Date") +
  ylab("CPI Index") +
  theme_minimal()
```

